



UNIVERSITY
of HULL

Computer Science

BSc / MEng Computer Science for Games Programming

October 2026

Computer Science for Games Programming

Shaded modules are offered as options on your programme, all other modules are compulsory.

Level 4 (Certificate Stage)

Trimester 1	Trimester 2
Programming Portfolio (40cr. core)	
Algorithms and Data Structures	Databases and the Cloud
Professional Development	Computational Thinking

Common first year allows for easy transfer between the available programmes.

Level 5 (Diploma Stage)

Trimester 1	Trimester 2
Design, Develop, Deploy (40cr. core)	
Graphics Programming	Advanced Programming
Artificial Intelligence	eXtended Reality Development

Post-Diploma Stage (optional)

Trimester 1	Trimester 2
Year in Industry/Year Abroad	

Level 6 (Honours Stage)

Trimester 1	Trimester 2
Honours Stage Project (40 credits)	
Communicating and Teaching Computing (40 credits)	
Simulation and Artificial Intelligence for Games Programming	Parallel and Concurrent Programming
Games Architecture	Generative and Collaborative AI

Level 7 (Masters Stage) (optional)

Trimester 1	Trimester 2
Commercial Development Practice (40 credits)	
Real-time Graphics	Cutting Edge Concepts
C++ Programming and Design	Simulation and Concurrency

Please note that the content in the following pages is subject to change, as our curriculum evolves to match the rapidly changing Computer Science landscape.

Level 4

Programming Portfolio

Core tools: C#

Whether you've never written a line of code or you've been tinkering for years, this module meets you where you are. You'll learn to think like a programmer — breaking problems down, writing clean logic, and building working software in C#. By the end of the year you'll have moved from your very first program to designing with classes, objects and inheritance.

You'll explore:

- Core programming building blocks — variables, loops, functions and data types
- Searching, sorting and working with arrays, stacks and queues
- Object-oriented programming — classes, inheritance and encapsulation
- Reading, writing and managing code like a professional, using source control

Algorithms and Data Structures

Every great piece of software starts with a great idea for how to solve the problem — that's what this module teaches you to find. You'll step away from the screen and learn to think in code before you write it, exploring the algorithms and data structures that separate software that merely works from software that works brilliantly.

You'll explore:

- Pseudocode and state machines for planning before you program
- Core data structures, and how to choose the right one for the job
- Sorting and searching algorithms, and how to measure their efficiency
- Design thinking that carries into every other module on your degree

Databases and the Cloud

Core tools: C#, Microsoft SQL Server

Modern software rarely stores its data on just one machine anymore — this module shows you how databases and the cloud work together to make that possible. You'll learn how cloud computing and virtualisation actually work, then design and build real databases, from relational systems through to NoSQL, before bringing the two together to build a working application backed by a database hosted in the cloud.

You'll explore:

- Designing and implementing relational and NoSQL databases
- Virtualisation, containers, and the cloud service models behind them — IaaS, PaaS and SaaS
- Connecting a real application to a cloud-hosted database

- Cloud security, reliability and disaster recover

Computational Thinking

Core tools: programming platforms and mathematical software

Computing is bigger than any one programming language — this module is about how to think as a computer scientist. Drawing on a genuinely global history of innovation, you'll build the mathematical and logical foundations that everything else on your degree rests on.

You'll explore:

- Abstraction and computational methods
- The maths behind computing — algebra, number systems and statistics
- Logic, Boolean algebra and set theory
- Vectors and matrices, and their uses in graphics, AI and data
- Practical aspects of how computers handle different number types and the way that mathematical libraries and tools can help to do the actual calculations

Professional Development

A computer science degree is about more than code — this module helps you build the career to go with it. You'll explore the ethical, legal, social and sustainability challenges facing the sector, and the professional skills that set graduates apart. A hands-on team project puts those skills into practice straight away, while you start building a portfolio that showcases your work.

You'll explore:

- Sector challenges — ethics, diversity, sustainability and cybersecurity
- Communication, research and self-presentation skills
- Real teamwork on a programme-specific group project
- Building your own professional portfolio

Level 5

Design, Develop, Deploy

Core tools: *GitHub*, *GitHub Actions*

This is where you learn to build software the way the industry actually builds it — from first idea to live deployment. As the backbone module of your second year, you'll work through the entire software development lifecycle, solo and as part of a team, picking up the methodologies and tools employers expect you to know.

You'll explore:

- Agile, iterative and waterfall development approaches
- Gathering and modelling real-world requirements
- Test-driven and behaviour-driven development
- Version control, CI/CD and cloud deployment with GitHub Actions

Advanced Programming

Core tools: C++

C++ powers everything from games engines to spacecraft — this module shows you why, from the inside out. You'll push beyond the basics of object-oriented programming and get close to the machine, exploring what's really happening between your code and the processor.

You'll explore:

- Templates and the Standard Template Library
- Composition and inheritance in advanced data structures
- Pointers, memory management and the “gang of three”
- Assembly language and debugging at the processor level

Artificial Intelligence

Learn to build the kind of intelligent systems that are reshaping every industry and understand the ethics behind them. You'll get a practical grounding in how AI actually works, from the history of the field to the algorithms powering it today, building your own solutions to real problem-solving challenges along the way.

You'll explore:

- The history and ethics and trustworthiness of artificial intelligence
- Pathfinding and problem-solving search algorithms
- Supervised and unsupervised learning, and neural networks
- Reasoning under uncertainty with Bayesian methods, and an introduction to reinforcement learning
- A brief introduction of fairness, explainability and AI Safety

Graphics Programming

Core tools: Shaders

Ever wondered how a game engine turns numbers into the world on your screen? This module shows you. Using a professional, industry-standard game engine, you'll learn the mathematics and pipeline behind real-time 3D graphics, and get hands-on writing your own shaders.

You'll explore:

- The evolution from fixed-function to programmable graphics pipelines
- Coordinate systems, vectors, matrices and transformations
- Building and animating 3D models and camera systems
- Writing custom shaders and using frame debugging tools

Year in Industry/Year Abroad

Take everything you've learned so far and put it to work — in a real workplace or a different country altogether. Spend a year gaining genuine professional experience on an industry

placement, or broaden your horizons studying at an overseas institution. Either route builds transferable, employability-boosting skills in a credit-bearing year that counts towards your degree.

You'll explore:

- A year of real workplace experience, tailored to your placement
- Or a year studying abroad, guided by department staff to match your goals

Level 6

Honours Stage Project

This is your chance to lead a project entirely your own — from first idea to final report. You'll choose your own topic and see it through from planning to delivery, supported by videos, live sessions and workshops covering everything you need to succeed.

You'll explore:

- Project planning and literature review
- Experimental design, testing and evaluation
- Research ethics and data protection
- Professional report writing and presentation

Communicating and Teaching Computing

Think you understand computer science? Try explaining it to someone else — that's the real test. Spend real time in schools or colleges as a computing ambassador, supported by workshops on teaching practice and the National Curriculum, and discover how explaining your subject deepens your own understanding of it.

You'll explore:

- Classroom management and lesson planning
- How computer science is taught in schools today
- Reflective, experience-based learning from your own placement
- Effective communication skills – a critical life skill whatever career you choose

Generative and Collaborative AI

AI isn't going to replace you — but someone who knows how to work with it might. This module makes sure that's you. You'll build three essential “fusion skills” for working alongside generative AI — asking it the right questions, judging its outputs critically, and teaching it to improve — then apply them to a real project of your own design.

You'll explore:

- Prompting techniques, from chain-of-thought to few-shot learning
- Critically evaluating AI-generated outputs for bias and accuracy
- Ethical, responsible use of generative AI

- A final project building a real generative AI solution

Games Architecture

Behind every great game is a great engine — learn how to build one. You'll explore the software architectures that power modern game engines, from component-based systems like Entity-Component-System through to the hardware considerations that shape how games run on console, PC and mobile.

You'll explore:

- Comparing and evaluating game engine architectures
- Entity-Component-System and modular design
- CPU architecture, bottlenecks and performance
- The asset pipeline and platform considerations

Parallel and Concurrent Programming

Core tools: CUDA, Rust

One processor is never enough — learn to make hundreds of them work together. You'll get hands-on with parallel programming on both CPU and GPU, implementing real algorithms in two very different languages to understand the challenges and rewards of true concurrency.

You'll explore:

- Shared memory vs message passing architectures
- Task, data and implicit parallelism
- Threads, channels and asynchronous programming
- Synchronisation, and avoiding the pitfalls of concurrency

Simulation and Artificial Intelligence for Games Programming

Give game characters a mind of their own — this module teaches you how. You'll write the data-driven code that brings games to life, building physics simulations and AI systems used in real production pipelines.

You'll explore:

- Physics-based simulation and collision response
- Pathfinding with Dijkstra maps and A*
- Steering behaviours, state machines and behaviour trees
- Advanced topics like flocking and procedural generation

Level 7

Commercial Development Practice

Build real software for a real client with industry-standard project management techniques and commercial tools. In Commercial Development Practice, the School's own commercial

software development unit, you'll work on genuine client projects that build the business and technical skills employers tell us are actively looking for.

You'll explore:

- Requirements analysis, defining specifications and system design
- Team-based implementation and quality control
- Client liaison, negotiation and project management
- Professional, reflective practice

C++ Programming and Design

Core tools: C++, Assembly

Go deeper into C++ than most programmers ever will — right down to the processor. Across four core areas — language fundamentals, object design, memory management and low-level execution — you'll master the features that make C++ the language of choice for performance-critical software.

You'll explore:

- Inheritance, aggregation and virtual functions
- Smart pointers, dynamic memory and exception safety
- Assembly language and processor-level debugging
- Code optimisation techniques

Cutting Edge Concepts

The textbooks can't keep up with computer science — so this module looks at what's happening right now. You'll work with staff to explore genuinely current research across AI, games, security and beyond, learning to critically evaluate academic sources and separate rigorous research from the rest.

You'll explore:

- Identifying high-quality academic sources
- Critically reviewing current research
- Exploring topics beyond the core curriculum
- Skills that carry straight into postgraduate study or research careers
- **Real-time Graphics**

Core tools: Vulkan, GLSL

Learn to render in real time at the level used in AAA games and film. You'll master the modern graphics pipeline on real shader-programmable hardware, implementing the lighting, shading and rendering techniques used across the games and animation industries.

You'll explore:

- Parallel processing models in 3D graphics

- The real-time graphics pipeline, hands-on
- Lighting, shading, texturing and shadow generation
- Multi-pass rendering and deferred shading

Simulation and Concurrency

Core tools: C++, Compute Shaders, Sockets

Simulate real physics, run it in parallel, and share it across a network — all in one module. You'll build real-time physical simulations, explore concurrent programming across CPU and GPU, and implement networked, distributed solutions using sockets.

You'll explore:

- Newtonian mechanics and collision response
- Parallel programming with threading and GPU concurrency
- Lock-based and lockless synchronisation
- Networked, distributed application design